THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 6 — SCIENCE · HARD LEVEL UNIT TEST SET 2

Chapters 1–4: **The Wonderful World of Science, Diversity in the Living World, Mindful Eating: A Path to a Healthy Body, Exploring Magnets** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. A freely hung bar magnet has one end painted red; the red end settles pointing towards the geographic south. The red end is the: 1

(a) south-seeking pole **(b)** neutral centre
(c) north-seeking pole **(d)** non-magnetic part

2. Plants A, B, C are given 100 mL, 200 mL, and 300 mL of water in same soil and light, and A grows least. Which is the changed variable? 1

(a) type of plant **(b)** amount of water **(c)** amount of light **(d)** amount of soil

3. Why do prey animals like deer have eyes on the sides of the head? 1

(a) to swim **(b)** wide field of view to spot predators
(c) to dig **(d)** to see only forward

4. A person eats only steamed vegetables and fruits and nothing else. The likely lack is: 1

(a) water **(b)** protein and fat **(c)** fibre **(d)** vitamins

5. A mixture contains aluminium, iron and wooden pieces. A magnet separates: 1

(a) iron only **(b)** wood
(c) all metals **(d)** iron and aluminium

6. Which of the following is not good practice while recording data? 1

(a) rounding off numbers to look neat without noting it
(b) writing units
(c) noting the date
(d) giving a title

7. A jungle has more kinds of organisms than a desert because: 1

(a) it is colder **(b)** it provides more food, water and shelter
(c) it has no animals **(d)** it has no plants

8. An experiment gave these results: Food X – blue-black with iodine; Food Y – oily patch; Food Z – no change in either test. Which is correct? 1

(a) X has starch, Y has fat, Z has neither starch nor fat
(b) X has fat, Y has starch, Z has both
(c) all three contain starch
(d) X has no starch, Y has starch, Z has fat

9. A student suspends magnet A and brings magnet B's north pole near A's north-seeking end. A turns away. This shows: 1

(a) A is iron **(b)** B's north repels A's north
(c) B's north attracts A's north **(d)** B has no poles

10. A scientist is convinced that her hypothesis is right, but the data shows otherwise. The correct action is to: 1

(a) repeat until the data matches **(b)** discard the data
(c) revise the hypothesis **(d)** change the readings

11. Which statement explains why camels have broad feet? 1

(a) to avoid sinking in sand **(b)** to dig for water
(c) to climb **(d)** to swim

12. A child gets tired quickly, has pale skin and breathlessness. Which food should the doctor suggest? 1

(a) chips **(b)** butter
(c) spinach and jaggery **(d)** sugar

13. Two magnets, each with poles marked, are placed in a line with N of A facing S of B and they attract. If B is reversed, they will: 1

(a) lose their magnetism **(b)** repel
(c) attract more strongly **(d)** attract as before

14. In the above experiment, which variable is measured as the outcome? 1

(a) growth of the plant **(b)** amount of water
(c) soil **(d)** sunlight

15. Which behaviour is an adaptation to a hot desert? 1

(a) being active at noon **(b)** swimming
(c) being active at night **(d)** hibernating in snow

SECTION — B (Very Short Answer Type) 5 × 2 = 10

Answer in about 20–30 words each.

16. Why does a dolphin look like a fish but is classified as a mammal? 2

17. Why is a diet rich only in proteins and fats not balanced? 2

18. A student says that heating a magnet strengthens it. Do you agree? 2

19. Give one example in which technology causes a problem and how science may solve it. 2

20. Why do aquatic plants such as hydrilla have thin, flexible stems? 2

SECTION — C (Short Answer Type) 5 × 3 = 15

Answer in about 40–60 words each.

21. Explain how a loudspeaker uses a magnet. 3

22. Explain the term 'variable' and give examples of three variables in an experiment on dissolving sugar. 3

23. Explain with an example how animals show adaptation in feeding. 3

24. How does one choose healthy snacks when travelling? 3

OR

How would a magnet help sort iron nails from a mixture of iron nails and brass screws? 3

25. In a plant experiment, plants P (light, water), Q (dark, water), R (light, no water). State what each comparison tests. 3

OR

Compare the adaptations of a duck and a fish for movement in water. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

26. Design a survey and experiment to study the effect of music on plant growth. Name the control and variables. 5

OR

Explain how the loss of habitat leads to loss of biodiversity. Suggest five ways to conserve it. 5

27. Discuss how traditional Indian meals combine foods to give a balanced diet, using thali as an example. 5

OR

Describe an activity to show that the strength of a magnet is highest at the poles, and write observations and a conclusion. 5

END OF QUESTION PAPER — Total: 50 Marks

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